CRIMEVISIONAI

Project Planning

Folder 4 | Backlog and implementation roadmap | Prepared 23 October 2026

Project: CrimeVisionAI — Multimodal Forensic Evidence Analysis and Investigation Support System

Document status: Project-specific draft based on the implementation information currently available. Verify placeholders and run-dependent results before formal submission.

1. Planning basis

The following backlog reflects the known implementation baseline and remaining quality gaps. Team member names, formal sprint dates and actual story-point completion were not supplied, so assignments remain role-based and dates are expressed as relative phases rather than invented historical facts.

2. Product backlog

Epic / ID	User story / task	Priority	Estimate (SP)	Status / acceptance
E1 / US-01	As a reviewer, I can create/select a case and see its evidence.	High	3	Basic case APIs reported; verify CRUD behavior.
E1 / US-02	As a reviewer, I can upload supported evidence to a case.	High	3	Upload/storage reported; test invalid types and size limits.
E2 / US-03	As a reviewer, I can inspect a timestamped audio transcript.	High	5	Whisper pipeline reported; benchmark transcription quality.
E2 / US-04	As a reviewer, I can inspect image detections, scene labels and caption.	High	5	RT-DETR/CLIP/BLIP pipeline reported; evaluate false positives.
E2 / US-05	As a reviewer, I can inspect video-frame analysis with timestamps.	High	8	1 FPS and 180-second cap reported; verify edge cases.
E3 / US-06	As a reviewer, I can review candidate events chronologically.	High	5	Timeline capability reported; validate timestamp consistency.
E3 / US-07	As a reviewer, I can ask questions about case analysis.	Medium	5	Case chat reported; grounding fix and regression suite needed.
E4 / US-08	As a reviewer, I can export a structured PDF report.	Medium	5	Not confirmed complete; implement/verify.
E5 / US-09	As a maintainer, I can run repeatable functional and performance tests.	High	5	Testing plan needed; do not claim performance numbers without results.
E5 / US-10	As an administrator, I can safely deploy with debug disabled and secrets protected.	High	8	Production hardening not confirmed; future milestone.

3. Suggested sprint / phase schedule

Phase	Focus	Deliverables / exit criteria
Phase 1 — Baseline audit	Confirm current code and runnable environment	Reproducible setup, dependency list, route inventory, DB schema.
Phase 2 — Core correctness	Image → video → timeline/data consistency → fusion → Q&A	Regression cases pass; known omissions documented and fixed where feasible.
Phase 3 — Testing & reporting	Functional tests, controlled benchmarks, PDF report	Evidence-backed test matrix and report export verified.
Phase 4 — Submission hardening	README, architecture docs, demo rehearsal, security review	Clean setup validated; limitations and future work clearly disclosed.

4. Risks and mitigations

Risk	Impact	Mitigation
Model output error	Incorrect observation or unsupported narrative	Use known-answer test assets, confidence labels, source verification and human review.
Q&A grounding gaps	Relevant entities may be omitted or answers may overstate evidence	Add retrieval/grounding tests and show evidence references.
Hardware-dependent latency	Slow demo or timeouts	Pre-download models, use representative test files and record hardware.
API availability / quota	LLM-dependent tasks may fail	Handle failures clearly; document required key and fallback behavior.
Unverified deployment/security	Exposure of evidence or secrets	Keep local demo private; harden before any public deployment.

5. Team allocation

Suggested role allocation: Project coordinator — backlog, integration and submission; Backend/model lead — Flask routes and analysis pipelines; Frontend lead — dashboard, upload, result and timeline UI; Testing/documentation lead — reproducible test evidence and deliverables. These are role suggestions, not claims about named contributors. Replace with the actual team allocation before submission.